

RESEARCH PAPER

Journal of
Biogeography

WILEY

Modelling the effect of environmental temperatures, microhabitat and behavioural thermoregulation on predicted activity patterns in a desert lizard across its thermally diverse distribution

Adam F. Parlin | Paul J. Schaeffer | Tereza Jezkova 

Department of Biology, Miami University,
Oxford, OH, USA

Correspondence

Tereza Jezkova, Department of Biology,
Miami University, Oxford, OH, USA.
jezkovt@miamioh.edu

Handling Editor: Enrique Martínez-Meyer

Abstract

Aim: Temperature plays an important role in determining distributions of ectothermic species, yet many ectotherms have wide distributions across diverse environmental conditions. Our goal was to model regional differences in annual and seasonal activity patterns among populations of *Phrynosoma platyrhinos* across its climatically diverse distribution. We hypothesize that microhabitat availability and behavioural thermoregulation will not mitigate large-scale climatic differences across the species' distribution, resulting in alteration of activity patterns of populations.

Location: Western deserts of North America.

Taxon: Desert Horned Lizard (*Phrynosoma platyrhinos*).

Methods: We used biophysical models to (a) model microhabitat conditions at 31 localities across the species' range, (b) model body temperature profiles of a theoretical lizard using microhabitat conditions, physiological and thermal parameters and the ability of lizards to thermoregulate behaviourally and (c) predict year-round activity patterns of *P. platyrhinos* across its range, quantified as the percentage of time permissible for basking or foraging.

Results: The models identified that different populations of *P. platyrhinos* experience differences in predicted activity patterns, both daily and seasonally, with up to 67% more time suitable for foraging in the southern than northern deserts. Southern populations have more potential for year-round activity, but predicted activity is reduced during hot summer months. The northern populations have less time for activity overall because of low temperatures in the spring and fall. However, summer-predicted activity is greater than in the southern regions.

Main conclusions: We detected pronounced regional differences in the times available to *P. platyrhinos* for activity. We conclude that behaviour and microclimate conditions are not able to fully mitigate the effects of large-scale thermal variation, and that lizards compensate for climatic variation by shifting their daily and seasonal predicted activity patterns. These differences can be tied to inter-population variation in life history traits.

KEYWORDS

biophysical model, foraging, NicheMapR, *Phrynosoma platyrhinos*, regional comparison

1 | INTRODUCTION

Animal distributions are limited by many factors, including environmental variation which may promote or inhibit persistence. Furthermore, these factors operate across different spatial and temporal scales. Species survival requires suitable environmental conditions not only at broadscale climatic conditions but also at the microclimatic scale of the animal (Kearney & Porter, 2009). For ectotherms, a critical factor that possesses strict upper and lower bounds for survival as well as thresholds for physiological function is body temperature. Although ectotherms experience a wide range of ambient temperatures, body temperatures must remain within tolerable physiological limits for survival and persistence at any developmental stage (Kearney & Porter, 2017). Furthermore, ambient conditions must be suitable for egg laying and there must be sufficient time in which juveniles are able to feed prior to overwintering (Gotthard, 2001). Given the reliance on ambient thermal conditions for regulation of body temperature, these temperatures play an important role in determining occupancy and persistence for many ectothermic species. In spite of this, many ectotherms have wide distributions that occupy a suite of environmental conditions that vary both daily and seasonally.

For species that have wide distributions and utilize highly variable environments, behavioural thermoregulation allows many ectotherms to keep body temperatures within their tolerable limits, necessary for survival and reproduction (Kearney, Shine, & Porter, 2009; Sunday, Bates, & Dulvy, 2011). An important aspect of this is movement within microclimates at a fine spatial scale, suggesting that environmental conditions measured at large spatial or temporal scales may not represent critical microclimate conditions experienced by individuals (Sunday et al., 2011, 2014). These local habitats can provide microclimatic conditions that buffer organisms within otherwise challenging climates, for example, shade providing relief from hot summer temperatures or burrows providing shelter from freezing winter temperatures (Cloudsley-Thompson, 2012; Newbold & MacMahon, 2014). Behavioural adaptations may thus mitigate the effects of large variation in climatic conditions across species' ranges (Adolph, 1990; Smith & Ballinger, 2001).

Alternatively (or in addition to behavioural thermoregulation), individual populations may respond to different environmental conditions throughout local modifications of either phenology or via changes in thermal performance optima or alteration of the limits to tolerance (Ayrinhac et al., 2004; Campbell-Staton et al., 2017). As a result, populations spanning diverse environments may exhibit differences in a suite of physiological or life history traits when sampled across their range (Ragland & Kingsolver, 2008; Smith & Ballinger, 2001).

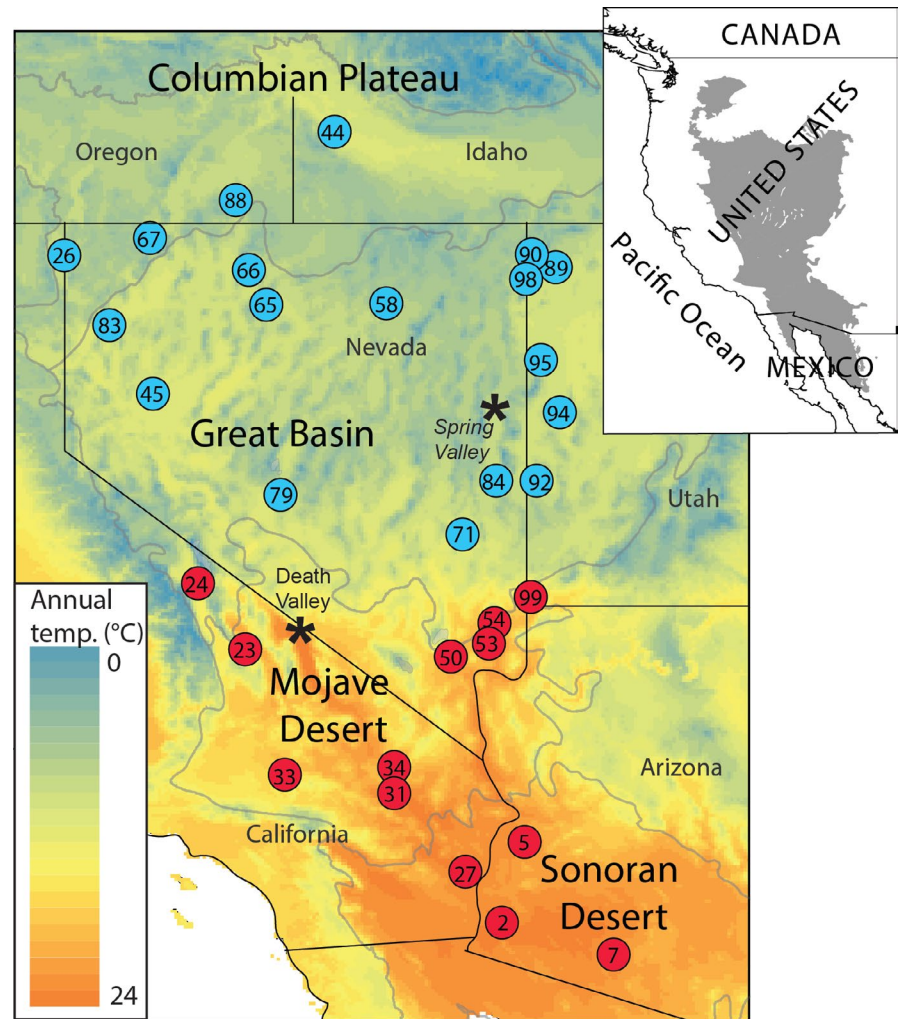
The Desert Horned Lizard (*Phrynosoma platyrhinos*) is an ectotherm that currently has a wide distribution, including some of the

hottest and driest places on Earth (e.g. Death Valley in the Mojave Desert) and cold desert conditions where freezing temperatures persist for up to 6 months of the year (e.g. Spring Valley in the Great Basin Desert; Figure 1). Yet, distinct populations found across this diverse range of climatic conditions have only been on separate evolutionary trajectories for ca. 10,000 years (Jezkova et al., 2016). This species inhabited only the warm Mojave and Sonoran Deserts during the Last Glacial Maximum (LGM). After post-LGM warming, it was able to persist within the Mojave Desert while facing warming of about 9°C (Betancourt, Van Devender, & Martin, 1990). Simultaneously, populations of this species expanded its range into the Great Basin Desert. Some parts of the Great Basin Desert that the species occupies today may be even colder than the climate *P. platyrhinos* experienced during the LGM (Jezkova et al., 2016). This species has therefore experienced rapid and substantial climatic changes throughout the last several thousand years, with populations that diverged from each other only recently now occupying habitats with very different climatic conditions. Previous work has shown regional differences in life history traits, such as clutch size and suggested seasonal variation in activity patterns, although activity data must be considered somewhat anecdotal (Pianka & Parker, 1975). These data serve as the inspiration for our predictive model, which seeks to explain mechanisms which could drive temporal and regional variation in predicted activity patterns of *P. platyrhinos*.

The complexity of the environment and resulting thermal landscape are important in determining how ectotherms cope with temperature extremes and how they use their habitat based on the thermal niches available (Sears & Angilletta, 2015). However, many models use coarse resolution thermal data (e.g. 1–4 km, WorldClim) to model the thermal landscape without incorporating the microclimate conditions available at the organismal scale (Kearney & Porter, 2017). Ectotherms may be capable of exploiting thermally suitable microhabitats that coarse resolution thermal data may fail to identify. Failure to include microclimate conditions and the daily thermal variation actually available can consequently lead to incorrect predictions of species' distribution, persistence and expansion (e.g. Jacobson et al., 2012). Biophysical ecology, using the laws of chemistry and physics for ecological interpretation (Gates, 1980), provides an approach for incorporating environmental variability, including solar and thermal radiation, convection, air temperature and relative humidity, to calculate microclimate conditions (Kearney & Porter, 2017). These models allow an ectotherm to optimize its body temperature behaviourally, which leads to predictions of biologically important activity patterns.

We use computational models based on biophysical relationships (a) to model microhabitat conditions, (b) to use these microhabitat values to calculate the body temperature and (c) to predict year-round daily activity patterns of a horned lizard across the range of

FIGURE 1 Occurrence sites ($n = 31$) of *Phrynosoma platyrhinos* complex used in this study (Appendix S2; Jezkova et al., 2016). Grey outlines (inset) represent the four main ecoregions of interest: Mojave Desert, Sonoran Desert (i.e. southern deserts), Great Basin, and Columbian Plateau (northern deserts; Olson et al. 2001). Red circles represent localities from the southern deserts where the species persisted during the Last Glacial Maximum. Blue circles represent localities from the northern deserts, into which the species expanded after the Last Glacial Maximum (Jezkova et al., 2016). The background shows the mean annual temperature (Hijmans, Cameron, Parra, Jones, & Jarvis, 2005). The asterisks mark two areas with the most extreme climatic differences within the species ranges, referred to in the text.



this species. The goal of our study was to characterize regional and temporal predicted activity patterns of populations of *P. platyrhinos* across its climatically diverse distribution to determine how these patterns are affected by environmental temperatures microhabitat and behavioural thermoregulation. We hypothesize that while microhabitat availability and behavioural thermoregulation buffer populations from lethal temperatures, these will not mitigate large-scale climatic differences in each region, resulting in alteration of annual and seasonal predicted activity patterns across the species' distribution.

2 | MATERIALS AND METHODS

2.1 | Study sites and organism

Desert ecoregions of western North America include the Mojave, the Sonoran, the Great Basin, and the Columbian Plateau (Figure 1). The Mojave and Sonoran Deserts are arid deserts with two distinct seasons. The winter months, between November and March, have moderate temperatures ranging from an average daily low of 3.5°C to an average daily high of 27.4°C. However, April–October

temperatures can reach as high as 46°C on average with average daily low temperatures of 16.4°C, potentially leading to 30°C temperature changes within a given day. Extreme temperatures average as high as 51.5°C and as low as 9.4°C between April and October, and as high as 34.2°C and as low as −1.9°C between November and March (WRCC, 2005; period of record: 06-11-1911 to 06-09-2016). To the north, the Great Basin Desert and the Columbian Plateau are temperate deserts with hot, dry summers and snowy winters. The winter months have an average daily low of −7.2°C and an average daily high of 9.3°C between November and March. During April–October, average temperatures are as high as 29.8°C and as low as −0.2°C. Extreme temperatures average as high as 34.2°C and as low as −7.7°C from April to October, and as high as 18.3°C and as low as −17.5°C between November and March (Western Regional Climate Center, 2005; period of record: 07-01-1948 to 03-31-2013).

The *P. platyrhinos* complex is currently widely distributed from the northwestern Sonoran and Mojave Deserts through the Great Basin Desert and Columbian Plateau (Figure 1). It comprises *P. platyrhinos* sensu stricto as well as closely related *P. goodei*, a lineage distributed in southeastern California, southern Arizona, and the adjacent state of Sonora. This lineage was elevated to a species status based on mitochondrial (mtDNA) sequence data and morphology

(Mulcahy, Spaulding, Mendelson, & Brodie, 2006). Genome-wide assessment, however, does not support genetic differentiation of *P. goodei* from *P. platyrhinos* (T. Jezkova, unpublished data). Herein, we use the name *P. platyrhinos* to refer to both *P. platyrhinos sensu stricto* and *P. goodei*.

To explore regional variation in predicted activity patterns in *P. platyrhinos*, we selected 31 representative sites from which *P. platyrhinos* has been previously collected for a genetic study (Jezkova et al., 2016) across the Mojave and northern Sonoran Deserts (for simplicity, we refer to these populations as southern) and east and west portions of the Great Basin Desert and southern edge of the Colombian Plateau (we refer to these populations as northern; Figure 1). The geographic coordinates assigned to each site correspond to an area of no more than 100 m² from which up to five lizards were collected. As described below, environmental data from each of the selected sites together with species-specific thermal thresholds were used in the computation of microclimate conditions, lizard body temperature and predicted activity patterns (model inputs described below).

Organismal data required for the biophysical model are parameters that determine behavioural patterns and transitions in predicted activity state. We used published data for the observed body temperature range of 23.8–40.2°C for active *P. platyrhinos* (Huey & Pianka, 2018), using the lower value for two behavioural thresholds, both temperature of emergence (T_{emerg}) and basking temperature (T_{bask}). Preferred temperature (T_{pref}) is the midpoint value of the range of selected temperatures recorded in a laboratory gradient (34.2°C; Lara-Reséndiz et al., 2015) and indicates the temperature at which the lizard will begin seeking shade to maintain body temperatures within the voluntary thermal limits. Thermal limits required for the biophysical model are voluntary thermal minimum (VT_{min}) and maximum (VT_{max}), which are the range of body temperatures a lizard will tolerate before becoming inactive through behavioural modifications. We based the voluntary thermal minimum (VT_{min} ; 31°C) for foraging on the minimum temperature that the primary food source for *P. platyrhinos* is active (Seed harvester ants, *Pogonomyrmex californicus*; Rissing, 1981). We chose this temperature as it is near the lower thermal preference range (*P. goodei*, 32.5°C; Lara-Reséndiz, Jezkova, Rosen, & Méndez-de La Cruz, 2014), the lower 25% quantile range of body temperatures observed in free-living horned lizards (33.4°C, Huey & Pianka, 2018), and because these lizards are ant specialists. We do not exclude the possibility that real lizards could choose to forage opportunistically below the VT_{min} threshold, but when the lizard's predicted temperature reaches 31°C, their food source becomes available. As we were unable to find a literature value for the voluntary upper thermal limit (VT_{max}), we determined this for lizards from both the Mojave and Great Basin regions following the methodology of Camacho et al. (2018). Briefly, individuals were collected in late May, transferred into the laboratory, held at room temperature with access to a heat source and processed within 48 hr of capture. VT_{max} was determined by placing each individual in a metal container placed inside a styrofoam box while

increasing the temperature inside. The lizards had an option of leaving the container when the temperature inside the container became unbearable. VT_{max} was recorded as the cloacal temperature of the lizard upon its voluntary exit ($n = 16$, across both regions). For this study, we assumed that the physiological parameters used in the model are similar across all sites. Measurement of VT_{max} complied with Miami University Institutional Animal Care and Use Committee protocol 992.

2.2 | Model application and analysis

Our overall approach was to (a) calculate microclimate and microhabitat conditions for each study site, (b) quantify the body temperature of a theoretical 30 g *P. platyrhinos*, (c) use these data to determine the heat budget model of the theoretical *P. platyrhinos* based on site-specific microclimate conditions and the ability of the lizard to move among microhabitats (sun, shade, burrow) and (d) determine predicted activity states of a lizard (i.e. inactivity, basking or foraging) based on thermal limits, each as described below.

Microclimate conditions are calculated for each locality (Figure 1) from weather, habitat and remote sensing data (see Kearney & Porter, 2019 supplemental for a complete list of parameters). Briefly, terrain is calculated using digital elevation data (Northwest Knowledge Network–THREDDS server) through the 'NicheMapR' v1.1.4 (Kearney & Porter, 2017) package to determine slope, aspect and elevation. Weather station variables were accessed from the GRIDMET daily weather database (<http://www.climatologylab.org/gridmet.html>) and specifically used precipitation, solar exposure, minimum and maximum relative humidity and minimum and maximum temperature. The habitat is generated from soil properties (Soil Grids; <http://rest.soilgrids.org/>) and shade conditions (minimum and maximum shade available). We determined the percentage of shaded area that *P. platyrhinos* could occupy. We first extracted the values of percentage of bare ground at each location from the Global 2010 Bare Ground raster with a resolution of ca. 30 m (Hansen et al., 2013). We then subtracted that percentage from 100 to get maximum potential shaded area that the model animal could select. We used these data in the 'micro_USA' function, which is based on the modified Fortran code of Porter, Mitchell, Beckman, and DeWitt (1973) that has been incorporated into 'R', to calculate microclimate conditions (which included: ground and air temperatures, relative humidity and temperature at animal's location). We based microclimate calculations at all 31 sampling locations on data from the 2017 calendar year and allowed burrowing (to 15 cm) and shade seeking behaviours for selection of thermally suitable conditions.

Microclimate data were then input into the 'ectotherm' function from the R package 'NicheMapR' v1.1.4 (Kearney & Porter, 2017) to quantify the body temperature of a theoretical 30 g *P. platyrhinos*. Calculated body temperature, combined with microclimate conditions and shade availability, was then used to determine the

predicted activity state of the animal using several temperature thresholds and thermal limits.

We determined lizard predicted activity states across 1 year as follows (Figure 2). The heat budget model attempts to maintain body temperature of the lizard within minimum (VT_{min}) and maximum (VT_{max}) voluntary thermal limits. The voluntary thermal limits indicate when the model animal has body temperatures that permit a foraging state, that is, above the VT_{min} but below the VT_{max} . The model begins by placing the theoretical lizard in an open environment at sunrise for each day, where it will remain as long as the ambient temperature is above the threshold for emergence temperature (T_{emerg}). If the calculated body temperature is greater than the basking temperature (T_{bask}) threshold, then the lizard model goes into a basking state. Note that we set both T_{emerg} and T_{bask} to the same temperature, thus we assume that once the lizard emerged, that it would begin a basking state unless body temperature was high enough to permit a foraging state. If the body temperature is greater than the voluntary thermal minimum (VT_{min}), the lizard model goes into a foraging state, which includes foraging, mate seeking or other activities. If the body temperature is greater than T_{pref} , then the lizard continues to be in a foraging state, but shade seeking occurs to maintain body temperatures within the voluntary thermal limits. Local shade availability may be low or high; thus, rates of heating or cooling may vary. If body temperature exceeds the voluntary thermal maximum (VT_{max}), the model lizard becomes inactive until conditions become thermally suitable, that is, body temperature is again within voluntary thermal limits of VT_{min} and VT_{max} . After sunset, if

still in a foraging state, the lizard becomes inactive. Predicted activity state values were output at hourly intervals.

2.3 | Data analysis

In order to compare the thermal limits and ensure consistency in the model parameters, we used a Student's t test to test for differences in voluntary thermal maximum (VT_{max}) between horned lizards from the Mojave and Great Basin Deserts. After running all the models, we then split the 2017 calendar year into three portions for comparisons between regions as follows. The beginning period (January–April) represents emergence and the energetically demanding mating period in which males seek mates and females gain body mass to provision eggs (Sinervo et al., 2010). The middle period (May–August) represents summer in which they lay eggs and must deal with heat loads. The end period (September–December) is important for juvenile persistence as they must gain sufficient mass to survive over-wintering (Pianka & Parker, 1975). We also analyse predicted activity across the entire year to assess the overall time available for biologically important functions. We imported monthly percentage of time basking, foraging and inactive for each portion of the year (beginning, middle and end) into ArcGIS, and interpolated them across uniformly spaced 2.5 arcmin grids, using the inverse distance weighted interpolation procedure (Watson, 1985) in the Spatial Analyst extension of ArcGIS. We then visualized the patterns of the three different time periods across landscapes focusing on the

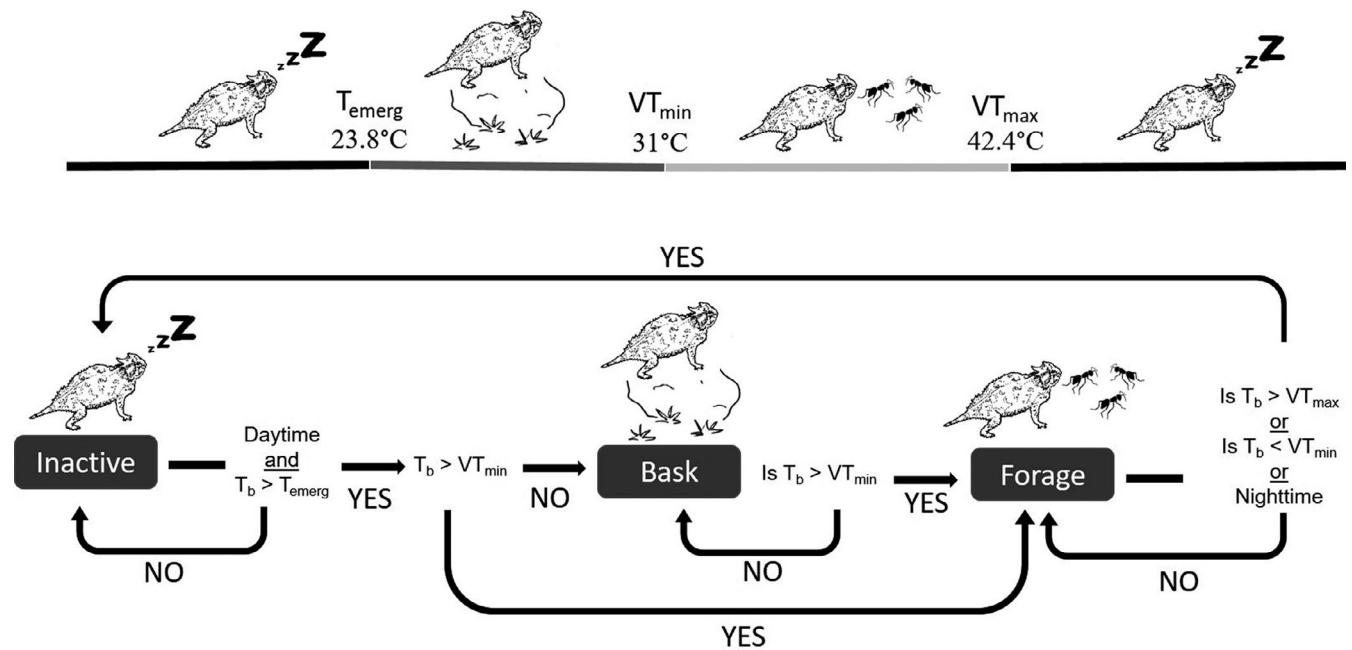


FIGURE 2 Determination of *Phrynosoma platyrhinos* activity state is based on the body temperature (T_b) of the lizard as determined by the model and compared with both day/night status and the values for several other temperature thresholds. The lizard is considered inactive if it is night-time or if T_b is either below the minimal temperature for burrow emergence (T_{emerg}) or above the experimentally determine maximal temperature that is voluntarily tolerated (VT_{max}). If daytime and T_b exceeds T_{emerg} , then the lizard begins basking to increase its T_b . T_{bask} spans T_{emerg} to VT_{min} . If T_b exceeds VT_{min} , then the lizard is determined to be foraging (which encompasses all active behaviours). Should T_b fall, the decision tree is enacted in reverse.

TABLE 1 Comparison of calculated body temperature (mean $\pm$ SD) and overall percentage of activity state (% $\pm$ SD) based on model output for each region, Great Basin and Mojave Deserts, for beginning (January–April), middle (May–August) and end (September–December) of the calendar year. East and West Great Basin were combined because of the similarity in microclimate conditions. Although limited by midday summer heat, lizards in the Mojave Desert had warmer body temperatures and more time throughout the year for *foraging* while lizards in the Great Basin Desert are limited to activity in the middle of the year due to cold conditions during the beginning and end of the year, leading to an overall reduction in annual activity. Post hoc comparisons (Tukey-adjusted least-squares means) are denoted with superscripts for foraging, basking and inactive comparing across the two major regions with letters indicating significant differences ($p < 0.05$)

Region	Season	Body temp. (°C)	Inactive (%)	Basking (%)	Foraging (%)
Great Basin	Beginning	7.07 $\pm$ 1.42 ^a	92.37 $\pm$ 3.01 ^e	4.59 $\pm$ 1.02 ^{a,b}	3.03 $\pm$ 2.31 ^a
	Middle	23.31 $\pm$ 1.05 ^b	62.03 $\pm$ 2.95 ^b	8.49 $\pm$ 0.47 ^{c,d}	29.48 $\pm$ 2.62 ^d
	End	11.41 $\pm$ 1.50 ^c	83.62 $\pm$ 3.98 ^d	6.51 $\pm$ 1.69 ^{c,d}	9.87 $\pm$ 2.58 ^b
Mojave	Beginning	15.44 $\pm$ 2.83 ^d	77.96 $\pm$ 3.81 ^d	7.29 $\pm$ 1.44 ^{a,c}	14.75 $\pm$ 2.52 ^b
	Middle	29.47 $\pm$ 2.09 ^e	74.46 $\pm$ 3.29 ^a	5.66 $\pm$ 1.23 ^d	19.88 $\pm$ 2.54 ^e
	End	19.03 $\pm$ 2.12 ^f	72.80 $\pm$ 1.89 ^c	7.56 $\pm$ 0.97 ^{b,d}	19.63 $\pm$ 1.68 ^c

beginning, middle, end and on the entire annual cycle from January 1 to December 31 2017.

To compare the among-site variation in the Mojave Desert, and Eastern and Western arms of the Great Basin Desert, we used principle component analysis (PCA) for percentage of time foraging (within VT_{min} and VT_{max}) and percentage of time basking (above T_{BASK} but below VT_{min}). Percentage of time for basking or foraging was determined by calculating the number of hours in either mode of predicted activity based on the model output. Given that all of the data were in the same units, we did not normalize prior to running the PCA. Principle components were considered significant if the PC-loading was greater than 0.5. Based on the PCA results, we also generated and compared the predicted activity plots between the two sites that were the most diverse in terms of percentage of time within basking and foraging for a standard calendar year from January 1 until December 31. PCA was performed using the base statistics package in R (R Core Team, 2016).

We analysed the thermal model output for estimated body temperature using generalized linear mixed models (GLMM) with the 'glmmTMB' R package (Brooks et al., 2017). We analysed the data in two ways: the first incorporated a complete 24-hr cycle, while the second focused on daylight hours when the zenith angle was less than 90°, indicating sunlight. For all models, we used *Region* and *Season* as our fixed effect and *Site* as our random effect with *Time of Day* as an autoregressive term within *Month* to correct for temporal autocorrelation. We compared models using a likelihood ratio test against a null model and delta AICc when necessary (Bolker, 2017), and then used least-squares means for a post hoc comparison when additional comparisons were necessary (Lenth, 2016). All model output data were analysed in the program R and all statistical analyses were tested for significance at the $\alpha = 0.05$ level.

Lastly, we compare the body temperature distribution from the model output with published data by Huey and Pianka (2018) for *P. platyrhinos* (see paper for details and supplemental code used to calculate skewness coefficient). We focus on a qualitative comparison of mean body temperature, median body temperature, mean

absolute deviation and D'Agostino coefficient of skewness as their measurements took place between 1962 and 1969 and our model output is based on climate from 2017.

3 | RESULTS

3.1 | Thermal tolerance and body temperature output

Initial thermal tolerance testing provided values for the model to calculate predicted activity based on body temperature. We found that there were no significant differences in the voluntary thermal maximum (VT_{max}) between *P. platyrhinos* in the Mojave or Great Basin regions ($t_{14} = 1.57$, $p = 0.13$, $n = 16$). The combined VT_{max} was 41.5 ± 2.0 (Mojave $VT_{max} = 42.85 \pm 1.42^\circ\text{C}$, Great Basin $VT_{max} = 40.23 \pm 1.52^\circ\text{C}$) and we use that value in the model. Although our sample size is not large, these results are consistent with other studies that suggest that VT_{max} remains conserved among closely related taxa (Grigg & Buckley, 2013).

First, we compared the model output for body temperature across regions (i.e. Mojave, East Great Basin and West Great Basin). We initially separated the two Great Basin subregions based on their range expansion history (Jezkova et al., 2016). We included the entire body temperature data set (day and night) to compare across the full variance in calculated body temperatures from the model. We found that *Region* was a more significant contributor to differences in the body temperature between sites ($X^2_7 = 45.81$, $p < 0.001$) than *Season* ($X^2_7 = 5.97$, $p = 0.050$) or the combination of *Region* + *Season* ($X^2_9 = 6.01$, $p = 0.049$) when tested against a null model (Appendix S1). Body temperatures generated by the model were compared using the least-squares means post hoc comparison of the regions using 95% confidence intervals (CI). These values were similar during the middle of the annual cycle comparing the East Great Basin (CI estimate: 9.14 ± 2.44) and the West Great Basin (CI estimate: 9.06 ± 2.44), but both were significantly different from the Mojave (CI estimate: 16.47 ± 2.42).

For temperatures of the model calculated only during daylight, all models were significantly different from the null model for *Region* ($X^2_7 = 45.8$, $p < 0.001$), *Season* ($X^2_7 = 8.23$, $p = 0.016$) and *Region + Season* ($X^2_9 = 8.29$, $p = 0.015$). Of the three competing models, the full model of *Region + Season* had the lowest delta AICc (Appendix S1). The least-squares means post hoc test indicated that daytime body temperatures in the middle of the annual cycle in the Mojave (CI estimate: 31.36 ± 3.49) were significantly greater than in the Eastern (CI estimate: 23.63 ± 3.51) and Western Great Basin (CI estimate: 22.38 ± 3.52).

Given that the two Great Basin subregions were similar, we present the combined values for the Great Basin, compared to the Mojave (Table 1). Using both day and night data, the model output body temperatures were higher on average in the Mojave Desert (CI_{beginning} estimate = 13.90–15.84; CI_{middle} estimate = 29.21–31.15; CI_{end} estimate = 17.93–19.87) compared to Great Basin Desert sites (CI_{beginning} estimate = 6.65–8.32; CI_{middle}

estimate = 21.97–23.63; CI_{end} estimate = 10.68–12.35), with similar variation (Table 1).

3.2 | Identification of inactivity, basking and foraging periods

Next, we analysed the time spent in each behavioural mode across region and season to determine the role of body temperature on predicted activity patterns. The percentage of time designated as foraging showed clear latitudinal separation between the Mojave and Great Basin deserts, ranging from 12% to 20% when averaged across the entire year (Figure 3a, Table S2). The percentage of time designated as foraging had greater variance among sites in the Great Basin Desert compared to the Mojave Desert (Table S2). When separated into each portion of the annual cycle, the Mojave and Great Basin Desert sites showed clear temporal differences (Table 1,

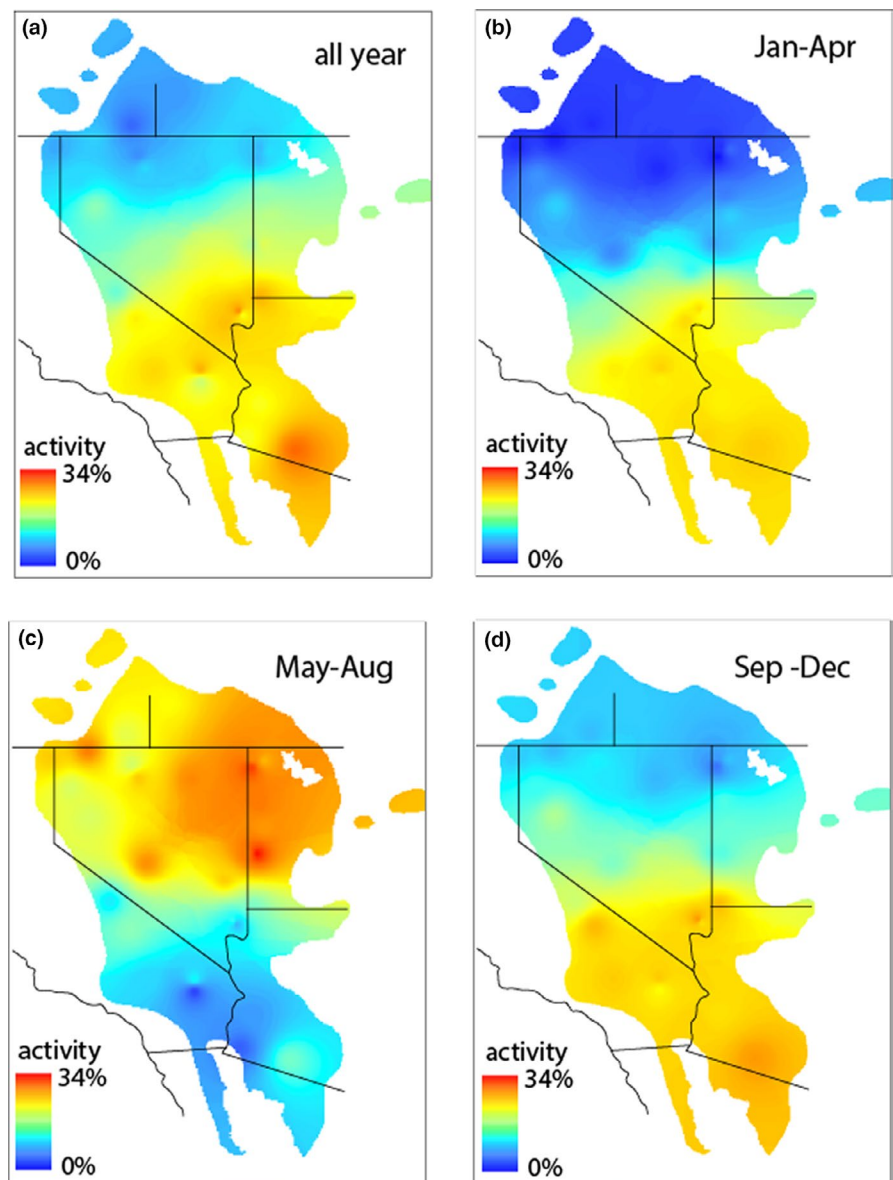


FIGURE 3 Geographic variation in activity patterns (% activity demonstrates the percentage of time designated as foraging and may include reproductive behaviour, foraging or other activities) throughout the year (a) and separately for the beginning (b), middle (c) and end (d) of the annual cycle

Figure 3b–d). During the beginning (Figure 3b) and end (Figure 3d) of the annual cycle, the percentage of time when body temperatures were within foraging was higher in the Mojave Desert than in the Great Basin Desert. For the Great Basin Desert, the middle of the annual cycle (Figure 3c) had the highest percentage of time with body temperatures within foraging.

We compared the Mojave Desert with East and West subregions of the Great Basin Desert to determine regional differences in time allocated to foraging and basking that may relate to the divergence in the range since the LGM using principal component analysis. The first two axes explained 97% of the total variation in the percentage of time body temperatures were within basking and foraging (Figure 4, Table 2). There was a clear separation between sites in the Mojave and both Great Basin subregions (East and West) in percentage of time within foraging using data for the whole year. The East and West Great Basin sites were similar in percentage of time available for basking and foraging with overlapping ellipses (95% CI) around the centroid (Figure 4). The first axis had negative PC-loading present for the beginning and end of the annual cycle for Mojave Desert sites, and a positive loading for the middle of the annual cycle for Great Basin Desert sites (Table 2). The second axis, which only

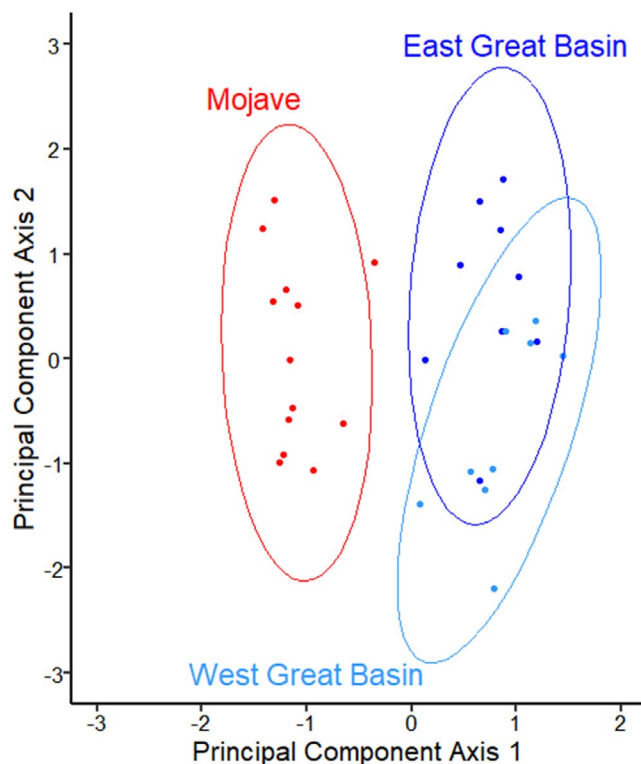


FIGURE 4 PCA results for percentage of time basking and foraging during the beginning, middle and end of the annual cycle. Mojave sites (red) were clustered based on negative loading from percentage of time foraging during the beginning and end of the annual cycle, while East Great Basin (dark blue) and West Great Basin (light blue) had a positive PC-loading from foraging during the middle of the annual cycle. Ellipses (95% CI) were drawn around each region to compare overlap and show a distinct separation between Mojave and Great Basin sites

explained 7% of the variation, only had a significant PC-loading for foraging during the middle of the annual cycle, explaining little of the separation between all sites. However, for both axes, foraging during the middle of the annual cycle had significant PC-loadings. Ellipses drawn around the centroid (95% CI) of the data sets shows no overlap between Mojave and Great Basin sites, indicating differences in percentage of time either basking or foraging. Designation as foraging was highest during the beginning and end of the annual cycle in the Mojave while for the Great Basin, it was highest during the middle of the annual cycle.

Two representative sites, one southern (Yuma—site 2) and the other northern (Wendover—site 98), demonstrate the daily pattern in percentage of time available for basking and foraging (Figure 5). We found that sites in the Mojave Desert and in the Great Basin varied in the amount of time within tolerable thermal limits for foraging, basking and inactive when comparing the beginning, middle and end of the year (Figure 5). *Region* and *Season* were significant contributors to the percentage of time in each predicted activity state compared to the null model (foraging: $X^2_3 = 103.17$, $p < 0.001$; basking: $X^2_3 = 15.58$, $p = 0.0014$; inactive: $X^2_3 = 85.04$, $p < 0.001$). Based on the model output, *P. platyrhinos* in the Mojave sites have more time within foraging during the beginning and end of the annual cycle than during the middle of the annual cycle when thermal conditions become too hostile and the model animal became inactive (see Table 1 for post hoc least-square means comparisons). Comparatively, the thermal conditions during the middle of the annual cycle for Great Basin sites allow for daily foraging activity with minimal basking (Table 1). However, at the beginning and end of the annual cycle, the model was unable to select thermally tolerable conditions and thus became inactive. Predicted inactivity in the Mojave site was due to thermal extremes being too hot during the summer, where body temperatures were above the voluntary thermal maximum (VT_{max}), while predicted inactivity in the Great Basin site was due to thermal extremes being too cold during the winter, where body temperatures were below the emergence temperature (T_{emerg}).

3.3 | Model output comparisons with published field body temperatures

We compare the basking and foraging activity state body temperatures of the model output with the data reported in Huey and Pianka (2018). For field collected *P. platyrhinos* measurements, the skewness was less than -1 for both northern and southern regions indicating a highly skewed distribution (North: mean $T_b = 34.3^\circ\text{C}$, median $T_b = 35.2^\circ\text{C}$, MAD = 2.67°C , skewness coefficient = -1.01 ; South: mean $T_b = 35.9^\circ\text{C}$, median $T_b = 36.6^\circ\text{C}$, MAD = 1.78°C , skewness coefficient = -1.58). Our model output based on 2017 climate data had a skewness coefficient between -0.5 and 0.5 indicating a symmetrical distribution (Great Basin: mean $T_b = 32.5^\circ\text{C}$, median $T_b = 33.5^\circ\text{C}$, MAD = 3.42°C , skewness coefficient = -0.17 ; Mojave: mean $T_b = 33.2^\circ\text{C}$, median $T_b = 33.7^\circ\text{C}$, MAD = 3.99°C , skewness coefficient = -0.25). We note that our model outputs have

continuous calculated body temperature measurements based on 2017 climate data while the published field-collected data required physical capture and were collected between 1962 and 1969.

4 | DISCUSSION

Phrynosoma platyrhinos is found across a thermally diverse range, the Mojave and Great Basin deserts, which suggested that activity across this range may vary accordingly. We present broadscale geographic comparisons of predicted activity patterns and their

potential phenological, behavioural and physiological mechanisms that are based on the use of biophysical models to determine body temperatures of an individual under varying macro and/or micro-climatic conditions. Comparison of model output with previously published field-collected data shows similarity in mean, median and mean absolute deviation. However, the natural populations experience highly skewed body temperature distributions and our models produced symmetrical temperature distributions, possibly an artefact of continuous calculation by the model and physical capture for the field collected data. Our results emphasize the utility of biophysical modelling to quantify daily and seasonal predicted activity of ectotherms.

We found that differences in the model output of predicted activity patterns were largely due to the thermal extremes experienced in each desert region, specifically the extreme high temperatures in the Mojave Desert during the summer and extreme low temperatures in the Great Basin Desert during the winter. Our results suggest that microclimatic selection and behavioural thermoregulation do not compensate for large-scale climatic differences across the species' range. The thermal conditions suitable for activity range from 20% of daytime in the southern deserts to 12% in some areas of the northern deserts across the year. The overall differences in suitable thermal conditions are further magnified during individual portions of the year. The southern populations show a bimodal seasonal increase in predicted activity for foraging, with higher activity in the beginning and end of the annual cycle and an inhibition of activity during the middle of the annual cycle due to high temperatures, with the exception of crepuscular activity. *Phrynosoma blainvillii* and *P. mcallii*, two species found in southern California, have been documented

TABLE 2 Component loadings of percentage of time for basking or foraging during the beginning, middle and end (PC1, PC2) of the annual cycle based on microclimate simulations for all 31 merged into regions (Mojave, East Great Basin and West Great Basin). Percentages of time with a positive or negative factor loading of at least 0.5 or -0.5 were considered to contribute to the component (indicated in bold)

	PC1	PC2
Basking—beginning	−0.15	0.01
Foraging—beginning	−0.63	0.28
Basking – middle	0.15	−0.14
Foraging—middle	0.51	−0.76
Basking—end	−0.06	0.40
Foraging—end	−0.53	0.38
Percent variance explained (%)	90.1	7.0
Total variance explained (%)	97.1	

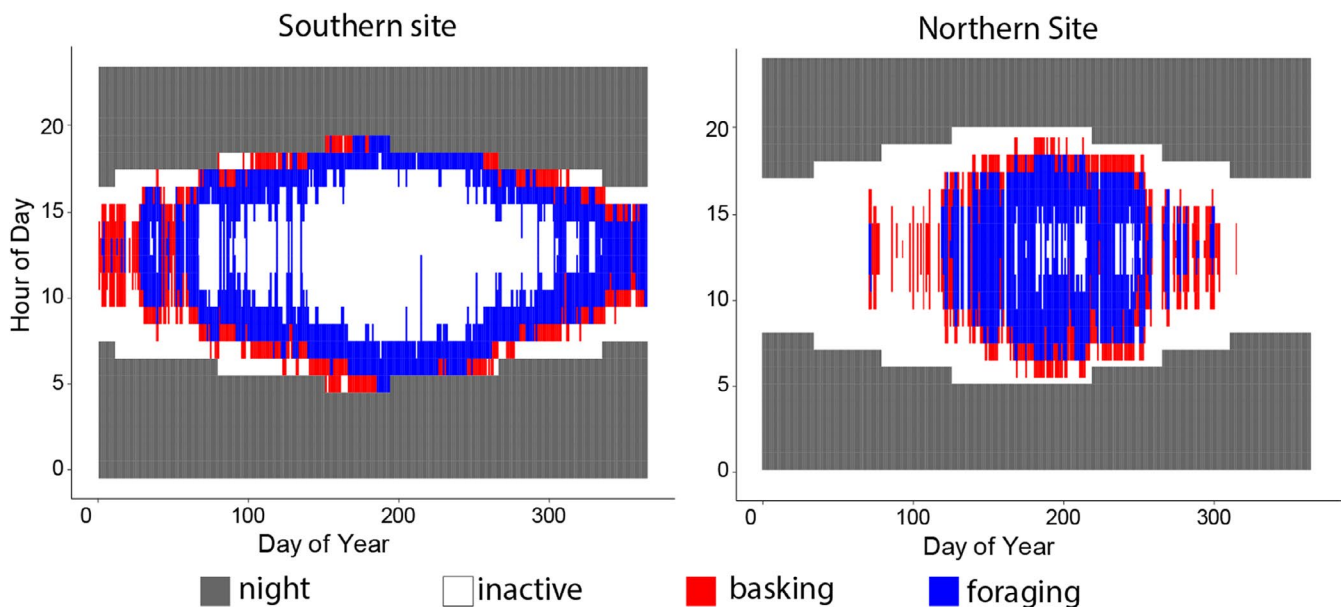


FIGURE 5 Activity plot for *Phrynosoma platyrhinos* in a representative southern site (left) and a northern site (right). The grey represents night whereas the remaining area is the daytime. Inactivity (white) is either due to conditions being too hot or too cool. The sites in the south had year-round activity with a noteworthy loss of midday activity throughout a large section of the summer period. Conversely, the northern site had activity limited to the middle of the year when thermal conditions were suitable, while during the winter months, thermal conditions became too cold for emergence, and therefore, the model remained inactive and buried. Activity plots for the remaining sites can be found in Appendix S3

to be active from February till November (occasionally all year round), and switch from unimodal spring activity to a bimodal summer activity pattern (Hager & Brattstrom, 1997; Heath, 1965; Hult & Germano, 2015; Wone & Beauchamp, 2003). Northern populations show low levels of predicted activity during beginning and end of the year due to colder temperatures and highest activity throughout the day in the middle of the year (summer). The lizards from the southern deserts also show more pronounced daily bimodal predicted activity patterns, with highest activity during early morning and late evening hours. These results are consistent with field observations of temporal and spatial shifts in activity patterns in *P. platyrhinos* (Pianka & Parker, 1975). The spatial shifts in predicted activity patterns, specifically the foraging output, suggest that individual populations of *P. platyrhinos* exhibit adjustments which might be tied to plasticity or adaptation in various life history traits. We elaborate below on the biogeographic comparisons as they relate to natural history that may allow *P. platyrhinos* persistence during suboptimal conditions.

4.1 | Regional differences in predicted activity patterns

During activity periods, lizards tend to maintain their body temperature within the same tolerable limits throughout their geographic range, which often spans diverse macroclimatic conditions (Adolph & Porter, 1993; Andrews, 1998; Horváthová et al., 2013; Huey & Pianka, 2018; Kiefer, Van Sluys, & Rocha, 2005). They do this by utilizing microclimatic conditions and behavioural thermoregulation that allow them to maintain body temperature using available habitat conditions. The macroclimatic conditions, however, can constrain the amount of time during which a lizard can maintain its body temperature and therefore remain active, because microclimatic conditions and thermoregulation cannot compensate for extreme ambient temperatures, hot or cold (Adolph & Porter, 1993). In order to maintain body temperature within preferable limits during activity, lizards will adjust time of activity, both daily and seasonally (Hager & Brattstrom, 1997; Heath, 1965). This results in variation in the timing of activity periods and differences in the cumulative annual activity time in different populations throughout the species' range (Sperry, Blouin-Demers, Carfagno, & Weatherhead, 2010; Tinkle, Dunham, & Congdon, 1993). These differences in thermoregulatory behaviour are believed to represent at least partly a phenotypically plastic trait in wild population of lizards (Refsnider et al., 2018).

Our models predicted all three types of activity adjustments (daily, seasonal and cumulative) in response to large-scale thermal variation across the regions. In the southern deserts, total potential predicted activity (i.e. basking or foraging) was greater than in the northern deserts, with up to 67% more time suitable for foraging and reproduction. Differences in predicted activity patterns were further magnified when we split the year into beginning, middle and end of an annual cycle. The limiting low temperatures in spring and fall in the northern deserts resulted in a large reduction in time suitable for foraging and reproduction in comparison with southern sites. Some

sites in the Great Basin did not have any suitable time during the first 4 months (January–April). Conversely, the limiting high summer temperatures in the southern deserts resulted in a greater than 100% higher predicted activity time for northern populations during the four summer months. Daily activity adjustments show the most extreme differentiation between the northern and southern sites with less than a 10% overlap in times suitable for foraging when comparing the climatically most different southern and northern sites.

4.2 | Regional differences of natural history

Variation in daily activity patterns, seasonal activity patterns and cumulative annual activity time may result in alteration of life history traits of ectotherms, via both plasticity and adaptation. Daily activity patterns have been tied to shifts in diets (Huey & Pianka, 1983) and body size (Meiri, 2008). Cumulative activity times have been shown to be negatively correlated with annual survival rate due to longer exposure to predation and positively correlated with growth rate and fecundity due to having more time to forage (Adolph & Porter, 1993; Niewiarowski, 2001; Sears, 2005). Body size in ectotherms does not show a consistent patterns as both increase and decrease of body size has been tied to climatic variation and variation in activity times (Ashton & Feldman, 2003; Caruso, Sears, Adams, & Lips, 2014; Iraeta, Monasterio, Salvador, & Díaz, 2006; Ramírez-Bautista, Leyte-Manrique, Marshall, & Smith, 2011; Sears, 2005). These seemingly contradicting patterns are likely tied to differential strategies of energy allocation with respect to growth, storage and reproduction (Grant & Dunham, 1990; Niewiarowski, 2001).

In *P. platyrhinos*, Pianka and Parker (1975) observed spatial and temporal difference in the number of clutches, and spatial difference in clutch size with larger clutch sizes reported in Great Basin populations. The northern populations often lay a single, larger clutch whereas southern females may regularly lay two clutches, each with a smaller number of eggs, a pattern found in other species of ectotherms (e.g. Niewiarowski, 2001; Tinkle, 1961). This is likely directly linked to the number of days with suitable conditions for egg laying, egg development and juvenile growth. Juvenile size is critical for successful overwintering and lizards hatched in the Great Basin Desert later in the season typically do not survive the winter (Pianka & Parker, 1975). *Phrynosoma platyrhinos* further exhibits spatial differences in male body size, with the northern males being smaller (Pianka & Parker, 1975). This size difference may be linked to lower survival of males in the northern populations (i.e. average male in the Great Basin is younger and therefore smaller) or due to more feeding days in the south (i.e. the growth rate is faster in the southern deserts).

We have no evidence that thermal tolerance and preference differ among populations of *P. platyrhinos*, but thermal characteristics of populations are likely to be subjected to adaptive pressures. These pressures may have caused shifts in thermo-physiological traits (such as temperature preference, critical maximum and minimum temperatures and optimal temperatures for locomotor

performance). Differences in thermo-physiological traits have been documented at the interspecific as well as intraspecific level. Lower thermal limits generally correlate with environmental conditions these species occupy (e.g. lower critical thermal minimum of species in cold regions; Campbell-Staton et al., 2017; Lara-Reséndiz et al., 2015). For example, if T_{emerg} or VT_{max} are lower in the northern than in southern populations, this would increase the time suitable to northern lizards for basking or foraging. Additionally, the severe and prolonged freezing temperatures in parts of the Great Basin suggest the existence of physiological mechanisms that allow lizards to survive the subfreezing temperatures, such as supercooling or freeze tolerance (Storey & Storey, 1992). Conversely, upper thermal limits (e.g. CT_{max} , VT_{max}) tend to stay conserved among related taxa (Grigg & Buckley, 2013), which are consistent with our findings of no differences in VT_{max} between the Mojave and Great Basin populations of *P. platyrhinos*. Therefore, we do not expect our models to overestimate times predicted as unsuitable due to high temperatures in the southern regions.

5 | CONCLUSIONS

Biogeographic comparisons at the scale of the organism is challenging given the climatic data needed. Furthermore, accurate modelling of variation at the microclimatic scale that influences daily activity and behaviour is essential. Our approach guides to a better understanding of what factors play an important role in the life history of an organism, whether physiological, behavioural or phenological. Biophysical models offer a technique for computationally determining the microclimate conditions and the subsequent effects on functional parameters of organisms. Given that biophysical models can be applied across wide landscapes with species-specific physiological and behavioural parameters, attempts at large-scale geographic comparisons are more feasible and can better inform subsequent experiments. Although the model shows when conditions are favourable for the species, actual horned lizards may differ from the predicted activity patterns calculated by the model. The body temperatures calculated in this study are based on biophysical principles that cannot completely incorporate the stochastic nature of organismal behaviour. This biophysical model serves as an important step towards understanding broad patterns activity as well as the mechanisms that underlie those patterns. Furthermore, it identifies the importance of microclimate temperature and indicates the value of further, experimentally driven studies of these parameters across biogeographic comparisons.

ACKNOWLEDGEMENTS

We thank D. Berg and J. Bracken for comments and suggestions.

DATA AVAILABILITY STATEMENT

Data available at Dryad <https://doi.org/10.5061/dryad.q2bvq83gj>.

ORCID

Tereza Jezkova  <https://orcid.org/0000-0003-4114-4564>

REFERENCES

- Adolph, S. C. (1990). Influence of behavioral thermoregulation on microhabitat use by two *Sceloporus* lizards. *Ecology*, 71(1), 315–327. <https://doi.org/10.2307/194027>
- Adolph, S. C., & Porter, W. P. (1993). Temperature, activity, and lizard life histories. *The American Naturalist*, 142(2), 273–295. <https://doi.org/10.1086/285538>
- Andrews, R. M. (1998). Geographic variation in field body temperature of *Sceloporus* lizards. *Journal of Thermal Biology*, 23(6), 329–334. [https://doi.org/10.1016/S0306-4565\(98\)00018-7](https://doi.org/10.1016/S0306-4565(98)00018-7)
- Ashton, K. G., & Feldman, C. R. (2003). Bergmann's rule in nonavian reptiles: Turtles follow it, lizards and snakes reverse it. *Evolution*, 57(5), 1151–1163. <https://doi.org/10.1111/j.0014-3820.2003.tb00324.x>
- Ayrinhac, A., Debat, V., Gibert, P., Kister, A., Legout, H., Moreteau, B., ... David, J. R. (2004). Cold adaptation in geographical populations of *Drosophila melanogaster*: Phenotypic plasticity is more important than genetic variability. *Functional Ecology*, 18(5), 700–706. <https://doi.org/10.1111/j.0269-8463.2004.00904.x>
- Betancourt, J. L., Van Devender, T. R., & Martin, P. S. (1990). *Packrat middens: The last 40,000 years of biotic change*, Tucson, AZ: University of Arizona Press.
- Bolker, B. M. (2017). glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *The R Journal*, 9(2), 378–400. <https://doi.org/10.3929/ethz-b-000240890>
- Brooks, M. E., Kristensen, K., Benthem, K. J., Magnusson, A., Berg, C. W., Nielsen, A., ... Bolker, B. M. (2017). glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. *The R Journal*, 9(2), 378–400. <https://doi.org/10.32614/RJ-2017-066>
- Camacho, A., Rusch, T., Ray, G., Telemeco, R. S., Rodrigues, M. T., & Angilletta, M. J. (2018). Measuring behavioral thermal tolerance to address hot topics in ecology, evolution, and conservation. *Journal of Thermal Biology*, 73, 71–79. <https://doi.org/10.1016/j.jtherbio.2018.01.009>
- Campbell-Staton, S. C., Cheviron, Z. A., Rochette, N., Catchen, J., Losos, J. B., & Edwards, S. V. (2017). Winter storms drive rapid phenotypic, regulatory, and genomic shifts in the green anole lizard. *Science*, 357(6350), 495–498. <https://doi.org/10.1126/science.aam5512>
- Caruso, N. M., Sears, M. W., Adams, D. C., & Lips, K. R. (2014). Widespread rapid reductions in body size of adult salamanders in response to climate change. *Global Change Biology*, 20(6), 1751–1759. <https://doi.org/10.1111/gcb.12550>
- Cloudsley-Thompson, J. L. (2012). *Ecophysiology of desert arthropods and reptiles*, New York: Springer Science & Business Media.
- Gates, D. M. (1980). *Biophysical ecology*. New York: Springer.
- Gotthard, K. (2001). Growth strategies of ectothermic animals in temperate environments. In D. Atkinson, & M. Thorndyke (Eds.), *Environment and animal development: Genes, life histories, and plasticity* (pp. 287–303). Oxford, UK: BIOS Scientific.
- Grant, B. W., & Dunham, A. E. (1990). Elevational covariation in environmental constraints and life histories of the desert lizard *Sceloporus merriami*. *Ecology*, 71(5), 1765–1776. <https://doi.org/10.2307/1937584>
- Grigg, J. W., & Buckley, L. B. (2013). Conservatism of lizard thermal tolerances and body temperatures across evolutionary history and geography. *Biology Letters*, 9(2), 20121056. <https://doi.org/10.1098/rsbl.2012.1056>
- Hager, S. B., & Brattstrom, B. H. (1997). Surface activity of the San Diego horned lizard *Phrynosoma coronatum blainvillhi*. *The Southwestern Naturalist*, 42(3), 339–344. <https://www.jstor.org/stable/30055288>

- Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., ... Townshend, J. R. G. (2013). High-resolution global maps of 21st-century forest cover change. *Science*, 342(6160), 850–853. <https://doi.org/10.1126/science.1244693>
- Heath, J. E. (1965). *Temperature regulation and diurnal activity in horned lizards*. University of California Press. AUTHOR: Please provide publisher location for reference Heath (1965).
- Hijmans, R. J., Cameron, S. E., Parra, J. L., Jones, P. G., & Jarvis, A. (2005). Very high resolution interpolated climate surfaces for global land areas. *International Journal of Climatology*, 25(15), 1965–1978. <https://doi.org/10.1002/joc.1276>
- Horvátová, T., Cooney, C. R., Fitze, P. S., Oksanen, T. A., Jelić, D., Ghira, I., ... Jandzik, D. (2013). Length of activity season drives geographic variation in body size of a widely distributed lizard. *Ecology and Evolution*, 3(8), 2424–2442. <https://doi.org/10.1002/ece3.613>
- Huey, R. B., & Pianka, E. R. (1983). *Lizard ecology*. Cambridge, MA: Harvard University Press.
- Huey, R. B., & Pianka, E. R. (2018). Body temperature distributions of active diurnal lizards in three deserts: Skewed up or skewed down? *Functional Ecology*, 32(2), 334–344. <https://doi.org/10.1111/1365-2435.12966>
- Hult, S. M., & Germano, D. J. (2015). Population structure, size, and activity patterns of *Phrynosoma blainvillii* in the San Joaquin Desert of California. *Herpetological Conservation and Biology*, 10(3), 839–849.
- Iraeta, P., Monasterio, C., Salvador, A., & Díaz, J. A. (2006). Mediterranean hatchling lizards grow faster at higher altitude: A reciprocal transplant experiment. *Functional Ecology*, 20(5), 865–872. <https://doi.org/10.1111/j.1365-2435.2006.01162.x>
- Jacobson, E. R., Barker, D. G., Barker, T. M., Mauldin, R., Avery, M. L., Engeman, R., & Secor, S. (2012). Environmental temperatures, physiology and behavior limit the range expansion of invasive Burmese pythons in southeastern USA. *Integrative Zoology*, 7(3), 271–285. <https://doi.org/10.1111/j.1749-4877.2012.00306.x>
- Jezkova, T., Jaeger, J. R., Oláh-Hemmings, V., Jones, K. B., Lara-Reséndiz, R. A., Mulcahy, D. G., & Riddle, B. R. (2016). Range and niche shifts in response to past climate change in the desert horned lizard *Phrynosoma platyrhinos*. *Ecography*, 39(5), 437–448. <https://doi.org/10.1111/ecog.01464>
- Kearney, M. R., & Porter, W. (2009). Mechanistic niche modelling: Combining physiological and spatial data to predict species' ranges. *Ecology Letters*, 12(4), 334–350. <https://doi.org/10.1111/j.1461-0248.2008.01277.x>
- Kearney, M. R., & Porter, W. P. (2017). NicheMapR—an R package for biophysical modelling: The microclimate model. *Ecography*, 40(5), 664–674. <https://doi.org/10.1111/ecog.02360>
- Kearney, M. R., & Porter, W. P. (2019). NicheMapR – An R package for biophysical modelling: The ectotherm and Dynamic Energy Budget models. *Ecography*, 43(1), 85–96. <https://doi.org/10.1111/ecog.04680>
- Kearney, M. R., Shine, R., & Porter, W. P. (2009). The potential for behavioral thermoregulation to buffer “cold-blooded” animals against climate warming. *Proceedings of the National Academy of Sciences of the United States of America*, 106(10), 3835–3840. <https://doi.org/10.1073/pnas.0808913106>
- Kiefer, M. C., Van Sluys, M., & Rocha, C. F. D. (2005). Body temperatures of *Tropidurus torquatus* (Squamata, Tropiduridae) from coastal populations: Do body temperatures vary along their geographic range? *Journal of Thermal Biology*, 30(6), 449–456. <https://doi.org/10.1016/j.jtherbio.2005.05.004>
- Lara-Reséndiz, R. A., Arenas-Moreno, D. M., Beltrán-Sánchez, E., Gramajo, W., Verdugo-Molina, J., Sherbrooke, W. C., & Méndez-De la Cruz, F. R. (2015). Selected body temperature of nine species of Mexican horned lizards (*Phrynosoma*). *Revista Mexicana De Biodiversidad*, 86(1), 275–278. <https://doi.org/10.1016/j.jtherbio.2014.11.010>
- Lara-Reséndiz, R. A., Jezkova, T., Rosen, P. C., & Méndez-de La Cruz, F. R. (2014). Thermoregulation during the summer season in the Goode's horned lizard *Phrynosoma goodei* (Iguania: Phrynosomatidae) in Sonoran Desert. *Amphibia-Reptilia*, 35(2), 161–172. <https://doi.org/10.1163/15685381-00002938>
- Lenth, R. V. (2016). Least-squares means: The R package lsmeans. *Journal of Statistical Software*, 69(1), 1–33. <https://doi.org/10.18637/jss.v069.i01>
- Meiri, S. (2008). Evolution and ecology of lizard body sizes. *Global Ecology and Biogeography*, 17(6), 724–734. <https://doi.org/10.1111/j.1466-8238.2008.00414.x>
- Mulcahy, D. G., Spaulding, A. W., Mendelson, J. R. III, & Brodie, E. D. Jr (2006). Phylogeography of the flat-tailed horned lizard (*Phrynosoma mcallii*) and systematics of the *P. mcallii*-*platyrhinos* mtDNA complex. *Molecular Ecology*, 15(7), 1807–1826. <https://doi.org/10.1111/j.1365-294X.2006.02892.x>
- Newbold, T. S., & MacMahon, J. A. (2014). Determinants of habitat selection by desert horned lizards (*Phrynosoma platyrhinos*): The importance of abiotic factors associated with vegetation structure. *Journal of Herpetology*, 48(3), 306–316. <https://doi.org/10.1670/10-141>
- Niewiarowski, P. H. (2001). Energy budgets, growth rates, and thermal constraints: Toward an integrative approach to the study of life-history variation. *American Naturalist*, 157(4), 421–433. <https://doi.org/10.1086/319321>
- Olson, D. M., Dinerstein, E., Wikramanayake, E. D., Burgess, N. D., Powell, G. V. N., Underwood, E. C., ... Loucks, C. J. (2001). Terrestrial ecoregions of the world: A new map of life on earth. *BioScience*, 51, 933–938.
- Pianka, E. R., & Parker, W. S. (1975). Ecology of horned lizards: A review with special reference to *Phrynosoma platyrhinos*. *Copeia*, 1975(1), 141–162. <https://doi.org/10.2307/1442418>
- Porter, W. P., Mitchell, J. W., Beckman, W. A., & DeWitt, C. B. (1973). Behavioral implications of mechanistic ecology. *Oecologia*, 13(1), 1–54. <https://doi.org/10.1007/BF00379617>
- R Core Team. (2016). R: A language and environment for statistical computing. Version 3.3.2. Vienna, Austria: R Foundation for Statistical Computing. <https://doi.org/10.1017/CBO9781107415324.004>
- Ragland, G. J., & Kingsolver, J. G. (2008). Evolution of thermotolerance in seasonal environments: The effects of annual temperature variation and life-history timing in *Wyeomyia smithii*. *Evolution*, 62(6), 1345–1357. <https://doi.org/10.1111/j.1558-5646.2008.00367.x>
- Ramírez-Bautista, A., Leyte-Manrique, A., Marshall, J. C., & Smith, G. R. (2011). Effects of elevation on litter-size variation among lizard populations in the *Sceloporus grammicus* complex (Phrynosomatidae) in Mexico. *Western North American Naturalist*, 71(2), 215–222. <https://doi.org/10.3398/064.071.0208>
- Refsnider, J. M., Qian, S. S., Streby, H. M., Carter, S. E., Clifton, I. T., Siefker, A. D., & Vazquez, T. K. (2018). Reciprocally transplanted lizards along an elevational gradient match light environment use of local lizards via phenotypic plasticity. *Functional Ecology*, 32(5), 1227–1236. <https://doi.org/10.1111/1365-2435.13071>
- Rissing, S. W. (1981). Prey preferences in the desert horned lizard: Influence of prey foraging method and aggressive behavior. *Ecology*, 62(4), 1031–1040. <https://doi.org/10.2307/1937002>
- Sears, M. W. (2005). Geographic variation in the life history of the sagebrush lizard: The role of thermal constraints on activity. *Oecologia*, 143(1), 25–36. <https://doi.org/10.1007/s00442-004-1767-0>
- Sears, M. W., & Angilletta, M. J. Jr (2015). Costs and benefits of thermoregulation revisited: Both the heterogeneity and spatial structure of temperature drive energetic costs. *The American Naturalist*, 185(4), E94–E102. <https://doi.org/10.1086/680008>



- Sinervo, B., Mendez-de-la-Cruz, F., Miles, D. B., Heulin, B., Bastiaans, E., Villagran-Santa Cruz, M., ... Sites, J. W. (2010). Erosion of lizard diversity by climate change and altered thermal niches. *Science*, 328, 894–899. <https://doi.org/10.1126/science.1184695>
- Smith, G. R., & Ballinger, R. E. (2001). The ecological consequences of habitat and microhabitat use in lizards: A review. *Contemporary Herpetology*, 3, 1–37. <https://doi.org/10.17161/ch.vi1.11957>
- Sperry, J. H., Blouin-Demers, G., Carfagno, G. L. F., & Weatherhead, P. J. (2010). Latitudinal variation in seasonal activity and mortality in ratsnakes (*Elaphe obsoleta*). *Ecology*, 91(6), 1860–1866. <https://doi.org/10.1890/09-1154.1>
- Storey, K. B., & Storey, J. M. (1992). Natural freeze tolerance in ectothermic vertebrates. *Annual Review of Physiology*, 54(1), 619–637. <https://doi.org/10.1146/annurev.ph.54.030192.003155>
- Sunday, J. M., Bates, A. E., & Dulvy, N. K. (2011). Global analysis of thermal tolerance and latitude in ectotherms. *Proceedings of the Royal Society B: Biological Sciences*, 278(1713), 1823–1830. <https://doi.org/10.1098/rspb.2010.1295>
- Sunday, J. M., Bates, A. E., Kearney, M. R., Colwell, R. K., Dulvy, N. K., Longino, J. T., & Huey, R. B. (2014). Thermal-safety margins and the necessity of thermoregulatory behavior across latitude and elevation. *Proceedings of the National Academy of Sciences of the United States of America*, 111(15), 5610–5615. <https://doi.org/10.1073/pnas.1316145111>
- Tinkle, D. W. (1961). Geographic variation in reproduction, size, sex ratio and maturity of *Sternotherus odoratus* (Testudinata: Chelydridae). *Ecology*, 42, 68–76.
- Tinkle, D. W., Dunham, A. E., & Congdon, J. D. (1993). Life history and demographic variation in the lizard *Sceloporus graciosus*: A long-term study. *Ecology*, 74(8), 2413–2429. <https://doi.org/10.2307/1939592>
- Watson, D. F. (1985). A refinement of inverse distance weighted interpolation. *Geoprocessing*, 2, 315–327.
- Western Regional Climate Center. (2005). WRCC home page. Accessed 22 February 2019.
- Wone, B., & Beauchamp, B. (2003). Movement, home range, and activity patterns of the horned lizard, *Phrynosoma mcallii*. *Journal of Herpetology*, 37(4), 679–686. <https://doi.org/10.1670/267-01A>

BIOSKETCH

Adam Parlin was a PhD student at the Department of Biology at Miami University studying energetics of body temperature regulation. He is currently a post-doctoral fellow in the Department of Biology at the University of Cincinnati.

Paul J. Schaeffer is an associate professor at the Department of Biology at Miami University studying phenotypic plasticity of thermal and energetic physiology.

Tereza Jezkova is an assistant professor at the Department of Biology at Miami University studying species' responses to climatic changes on regional scales.

AP and TJ conceived the study, AP performed the analyses, AP, PS, and TJ wrote the manuscript.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

How to cite this article: ParlinAF, Schaeffer PJ, Jezkova T.

Modelling the effect of environmental temperatures, microhabitat and behavioural thermoregulation on predicted activity patterns in a desert lizard across its thermally diverse distribution. *J Biogeogr.* 2020;00:1–13. <https://doi.org/10.1111/jbi.13936>